Process Improvement of Casting Inspection Using Six Sigma DMAIC Methodology to Reduce Downstream Processing Waste.

Process Improvement Case Study

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Executive Summary

This project applies the DMAIC (Define, Measure, Analyze, Improve, Control) methodology to study the casting inspection process in a manufacturing plant, with the aim of reducing irreparable defects that escape inspection and reach downstream operations.

Over a 15-day observation period, 9000 components were tracked. Of these, 189 components with irreparable defects passed the casting inspection stage undetected and were only caught later, after powder coating and machining, during final inspection. This gave a baseline defect escape rate of 2.10%, a First Pass Yield of 97.90%, and 21,000 DPMO roughly a 3.5 sigma. The escapes caused an estimated ₹3,591 in direct processing loss and 10.24 wasted machining hours during the study period, which projects to about ₹71,820 and 205 hours annually if left unaddressed.

Root cause analysis (using 5-Why and a fishbone diagram) showed that the defects were not being missed because of casting quality issues, but because of gaps in the inspection process itself which are no standardized acceptance criteria, insufficient inspector training, more focus on inspection speed over accuracy, and weak supervisory follow-up.

Based on these findings, this report proposes a set of practical, low-cost improvements: standardized inspection criteria and checklists, structured operator training, stronger supervisor audits, and a KPI-based monitoring system to track FPY, defect escape rate, and DPMO going forward. As this is an academic case study, these recommendations were not implemented, but they offer a clear, ready-to-use action plan targeting a defect escape rate below 1.0% and a sigma level above 3.8.

Project Snapshot

Content	Value
Methodology	DMAIC
Observation Period	15 days
Components	9000
Defect Escape	189
Defect Escape Rate	2.10%
FPY	97.90%
DPMO	21000
Sigma Level	~3.5 sigma
COPQ	₹3,591
Annual Loss	₹71,820
Primary Root Cause	Focus on Quantity Over Quality.
Proposed Target	<1.0% Defect Escape Rate

DEFINE

Problem Statement

During a 15-day industrial observation period, irreparable casting defects escaped the casting inspection stage and entered downstream manufacturing operations. These components subsequently underwent powder coating and machining before being identified as permanent rejects during final inspection, requiring them to be remelted.

Although some casting defects identified during final inspection were repairable and required only rework, this study focuses exclusively on irreparable defects that resulted in complete processing waste.

The escape of these defective components resulted in:

- Complete loss of powder coating material and processing cost.
- Complete loss of machining cost and machine time.
- Unnecessary utilization of downstream manufacturing resources.
- Increased inspection effort during final quality inspection.

The project focuses on improving the effectiveness of the casting inspection process to reduce the escape of irreparable defects into downstream operations.

Project Goal

The objective of this study is to improve the effectiveness of the casting inspection process by reducing the escape of irreparable casting defects into downstream manufacturing operations through standardized inspection practices and improved inspection control.

Project Scope

In Scope

- Visual inspection of casting components
- Identification of irreparable casting defects
- Standardization of inspection criteria
- Development of inspection checklists
- Operator training recommendations
- Supervisor audit recommendations
- Improvement of inspection documentation

Out of Scope

- Casting process parameters
- Die design modifications
- Furnace operation
- Alloy composition
- Repair methods for reworkable defects

Process Flow

Casting

↓

Casting Inspection

↓

Powder Coating

↓

Machining

↓

Final Inspection

Initial Operational Observations

During shop-floor observation, several operational and behavioral issues were identified that could contribute to defect escapes.

- **Quantity-Oriented Inspection**

Inspectors primarily focused on inspecting a large number of components within the shift. Greater importance was given to inspection quantity than inspection quality, increasing the possibility of overlooking critical defects.

- **Dependence on Final Inspection**

A common perception existed that any defects missed during casting inspection would eventually be identified during final inspection. This reduced the sense of responsibility at the casting inspection stage.

- **Lack of Structured Training**

Newly assigned inspectors received limited formal training before performing inspection activities. Inspection decisions were largely based on individual experience and personal judgment.

- **Limited Supervisory Follow-up**

When casting defects were detected during final inspection, systematic investigation and preventive actions were not consistently initiated at the casting inspection stage. Similar inspection errors therefore continued to recur.

MEASURE

Data Collection

Production data was collected over a continuous observation period of **15 days** from the casting production line. The study focused only on **irreparable casting defects** that passed the casting inspection stage and were later identified as permanent rejects during the final inspection after powder coating and machining.

For this study, each component was considered to have **one defect opportunity**, corresponding to the escape of an irreparable casting defect from the casting inspection stage.

Production Data

Day No.	Total Components	Casting Reject	Percentage
Day 1	600	8	1.33
Day 2	600	12	2
Day 3	600	10	1.66
Day 4	600	14	2.33
Day 5	600	17	2.83
Day 6	600	9	1.5
Day 7	600	13	2.16
Day 8	600	11	1.83
Day 9	600	15	2.5
Day 10	600	16	2.66
Day 11	600	8	1.33
Day 12	600	14	2.33
Day 13	600	10	1.66
Day 14	600	15	2.5
Day 15	600	17	2.83

Parameter	Value
Observation Period	15 Days
Total Components Processed (N)	9,000 Parts
Irreparable Defect Escapes (D)	189 Parts
Opportunities per Part (O)	1
Defect Escape Rate	2.10%

Defect Classification

- **Severe Uneven Surface Finish**

These components exhibited excessive surface irregularities, roughness, or pronounced molding marks that exceeded the acceptable quality limits and could not be corrected through downstream processing.

- **Deep Shrinkage Cavities / Surface Porosity**

These defects appeared as visible cavities or porosity near critical stress regions of the component. Due to their location and severity, the affected components were classified as permanent rejects.

Only these two defect categories were included in this study because they resulted in complete rejection.

Data Limitation: Individual defect types were not tracked separately; rejection data was recorded as a combined count. Because of that, it was not possible to do a Pareto analysis by defect category for this study. It is recommended that future data collection record the count according to defect type, to support creating a Pareto analysis-based action plan.

Baseline Process Performance

To evaluate the existing performance of the casting inspection process, the following quality metrics were calculated.

- **First Pass Yield (FPY)**

First Pass Yield represents the percentage of components that completed the manufacturing process without becoming irreparable rejects.

FPY = ((N – D) / N) × 100

Where,

N = Total number of components processed

D = Number of irreparable defect escapes

FPY **=** ((9000 − 189) / 9000) × 100

= (8811 / 9000) × 100

= 97.90%

Therefore, the First Pass Yield of the process was **97.90%**.

- **Defects Per Million Opportunities (DPMO)**

Defects Per Million Opportunities (DPMO) is a standard Six Sigma metric used to measure process performance by expressing the number of defects per one million opportunities.

DPMO = (D / (N × O)) × 1,000,000

Where,

D = Number of defect escapes

N = Total number of components

O = Opportunities per component

DPMO = (189 / (9000 × 1)) × 1,000,000

= 0.021 × 1,000,000

= 21,000

Therefore, the baseline process performance was **21,000 DPMO**.

Quality Metrics	**Value**
First Pass Yield (FPY)	97.90%
Defects Per Million Opportunities (DPMO)	21,000
Sigma Level	~3.5 sigma

Cost of Poor Quality (COPQ)

The financial impact was calculated only for the **189 irreparable rejected components**. Since the rejected aluminium components were remelted within the foundry, raw material cost was excluded from the analysis. Only downstream processing costs were considered.

- **Powder Coating Loss**

Powder coating cost per component = ₹7

Total powder coating loss

= 189 × ₹7

= ₹1,323

- **Machining Cost Loss**

Machining cost per component = ₹12

Total machining loss

= 189 × ₹12

= ₹2,268

- **Total Direct Processing Loss**

Total Processing Loss

= ₹1,323 + ₹2,268

= ₹3,591 (for 15 days)

Cost used in the Cost of Poor Quality (COPQ) analysis (powder coating and machining costs per component) were obtained from the Production Planning & Control department, ensuring the financial estimates reflect actual plant costs.

- **Machining Capacity Loss**

Every defective component occupied machining capacity before being rejected.

Machining cycle time per component = 3 minutes 15 seconds

= 3.25 minutes

Total machining time wasted

= 189 × 3.25

= 614.25 minutes

= 10.24 hours (for 15 days)

Therefore, approximately **10.24 hours of machining capacity** were consumed processing components that ultimately became permanent rejects.

- **Expected Annualized Impact**

To understand the long-term effect of defect escapes, the observed 15-day data was projected over one year.

Assuming 20 production cycles of 15 days as 300 working days are considered in one year,

- **Expected Annual Processing Loss**

Annual Processing Loss

= ₹3,591 × 20

= ₹71,820 per year

- **Expected Annual Machining Capacity Loss**

Annual Capacity Loss

= 10.24 × 20

= 204.8 hours per year

Key Findings

The collected data showed that **189** irreparable casting defects escaped the initial inspection stage, resulting in a **2.10%** defect escape rate, **97.90%** First Pass Yield, and **21000 DPMO.** These defect escapes generated **₹3591** in direct downstream processing loss during the 15 day observation period and consumed **10.24 hours** of machining capacity.

These findings indicated that improving the effectiveness of the casting inspection process could reduce unnecessary downstream processing and improve resource utilization.

ANALYZE

Root Cause Analysis

Based on shop-floor observations, discussions with personnel, and analysis of the collected production data, it was found that the problem was primarily related to inspection practices rather than the casting process itself. The investigation indicated that human factors, inspection methods, quality standards, and supervisory practices contributed to the occurrence of defect escapes.

5 Why Analysis

Problem: Irreparable casting defects escaped the casting inspection stage and reached downstream operations.

Why 1:
Critical casting defects were not detected during visual inspection.

Why 2:
Inspectors focused more on inspecting a higher number of components than on carefully evaluating each component.

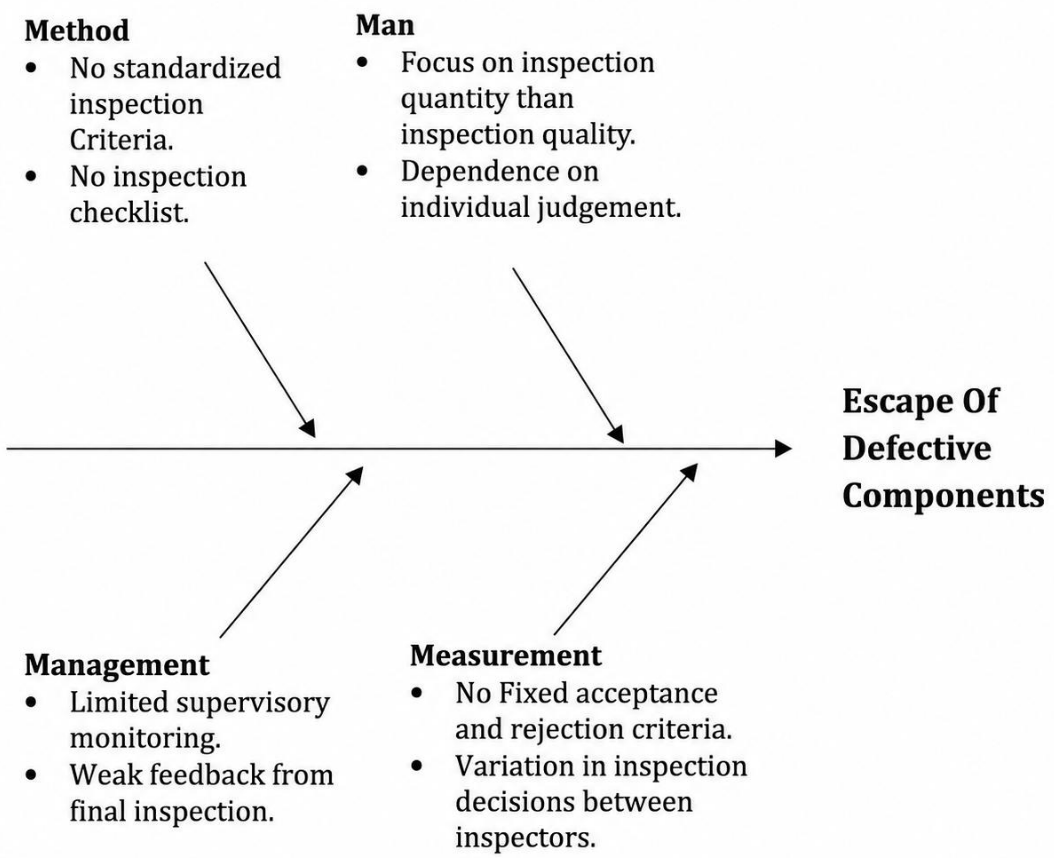
Why 3:
Inspection performance was largely judged by the quantity of components inspected rather than inspection accuracy.

Why 4:
Inspectors assumed that any defects missed during casting inspection would be identified during the final inspection stage.

Why 5:
There was no structured training program, standardized inspection procedure, or regular supervisory review to ensure consistent inspection practices.

The analysis indicated that the primary causes of defect escapes were associated with inconsistent inspection methods, lack of standardized quality criteria, insufficient training, and weak supervisory control.

Fish-Bone Diagram



IMPROVE

Recommended Improvement Actions

- **Standardize Visual Inspection Criteria**

It is recommended to establish standardized acceptance and rejection criteria for critical casting defects such as severe uneven surface finish and deep shrinkage cavities. Standardized visual reference charts with photographs of acceptable and non-acceptable defects may be displayed at each inspection station to support consistent inspection decisions.

- **Develop a Standard Operating Procedure (SOP)**

The organization could develop a Standard Operating Procedure (SOP) that clearly defines the inspection sequence, inspection method, defect identification process, and component acceptance or rejection criteria. A documented procedure may help reduce variation in inspection practices among different operators.

- **Introduce a Standardized Inspection Checklist**

A standardized inspection checklist may be introduced to ensure that every critical inspection point is verified before a component is approved for the next manufacturing stage. The checklist can help inspectors follow a consistent inspection sequence and reduce the possibility of missing important defects.

- **Conduct Structured Operator Training**

Training sessions should be conducted for both new and existing inspectors on casting defects, inspection procedures, and defect classification. Practical demonstrations using actual defective components may improve the ability of inspectors to identify irreparable defects consistently.
Periodic refresher training can also be arranged to maintain inspection competency.

- **Define Inspector Performance Criteria**

The organization could revise the existing performance evaluation system to include inspection quality in addition to inspection quantity. Performance indicators such as inspection accuracy, defect escape rate, and adherence to inspection procedures may be considered during performance evaluation.
This approach could encourage inspectors to prioritize inspection effectiveness rather than inspection speed alone.

- **Strengthen Supervisor Involvement**

Supervisors should regularly monitor inspection activities and verify that inspection procedures are followed correctly. Whenever defect escapes are identified during final inspection, supervisors should review the causes and communicate corrective actions to the casting inspection team.
Regular supervisory review may improve accountability and reduce repeated inspection errors.

- **Establish a Feedback System Between Inspection Stages**

A structured communication process can be established between the casting inspection and final inspection departments. Information related to recurring defect escapes should be shared regularly so that preventive actions can be planned at the source inspection stage.
This feedback system may improve coordination between departments and support continuous process improvement.

- **Review Defect Trends Periodically**

The organization could review defect data at regular intervals to identify recurring defect patterns and evaluate whether additional corrective actions are required. Trend analysis may assist management in monitoring the effectiveness of inspection practices over time.

Improvement Summary

Root Cause	**Recommended Improvement**
Subjective inspection decisions	Standardize visual inspection criteria and defect reference charts.
Undefined acceptance and rejection limits	Establish clear inspection standards and rejection boundaries.
Lack of standardized inspection process	Develop a Standard Operating Procedure (SOP).
No inspection checklist	Introduce a standardized inspection checklist.
Lack of operator training	Conduct structured training and periodic refresher sessions.
Quantity over quality mindset	Include inspection quality in performance evaluation.
Limited supervisory monitoring	Increase supervisor audits and inspection reviews.
Poor communication between departments	Establish regular feedback between casting inspection and final inspection.
Limited review of defect data	Conduct periodic analysis of defect trends.

Expected Benefits

If the proposed recommendations are implemented in the future, the organization may achieve the following benefits:

- Improved identification of irreparable casting defects during initial inspection.
- Reduced movement of defective components to downstream processes.

- Lower powder coating and machining processing losses.
- Better utilization of machining capacity.
- More consistent inspection decisions through standardized procedures.
- Improved inspector competency through structured training.
- Increased accountability through regular supervisory review.
- Better communication between inspection stages.
- Improved monitoring of process performance through regular review of defect trends.
- Enhanced process stability and support for continuous quality improvement.

The proposed actions focus on strengthening the inspection process rather than modifying the casting process itself, making them practical and suitable for implementation with minimal additional resources.

Targeted Improvement

Performance Matric	Current	Targeted
Defect Escape Rate	2.10%	<1.00%
FPY	97.90%	>99.00%
DPMO	21000	<10000
Sigma Level	3.5 Sigma	>3.8 Sigma
Processing Loss	₹71,820	<₹34200 (>52.38% Improvement)

CONTROL

Objective

The objective of the Control phase is to propose a practical control plan that could help sustain the recommended improvements and maintain consistent inspection performance if the proposed actions are implemented in the future. The control measures are intended to standardize inspection practices, monitor process performance, and reduce the recurrence of defect escapes.

Proposed Control Plan

Control Item	Proposed Control Method	Responsibility	Frequency	Record
Standard Operating Procedure (SOP)	Review compliance with the inspection SOP and update it whenever process changes occur.	Quality Supervisor	As Required	SOP Revision Record
Visual Inspection Standards	Display approved defect reference charts at each inspection station and verify their availability.	Quality Department	Continuous	Visual Standard Sheet
Inspection Checklist	Complete a standardized checklist before approving components for the next process.	Casting Inspector	Every Shift	Inspection Checklist
Operator Training	Conduct training for new inspectors and arrange refresher training for existing inspectors.	Production & Quality Department	Quarterly	Training Record

Supervisor Audit	Verify compliance with inspection procedures through regular shop-floor audits.	Production Supervisor	Weekly	Audit Report
Defect Tracking Sheet	Record all defect escapes identified during final inspection for trend analysis.	Quality Inspector	Daily	Defect Tracking Sheet
KPI Monitoring	Monitor First Pass Yield (FPY), Defect Escape Rate, and DPMO to evaluate inspection performance.	Quality Engineer	Monthly	KPI Report
Daily Review Meeting	Review previous day's inspection performance, recurring defects, and corrective actions.	Production & Quality Team	Daily	Meeting Minutes
Corrective Action Log	Record inspection-related issues and track corrective actions until completion.	Quality Department	As Required	Corrective Action Register
Process Documentation	Maintain inspection records, audit reports, training records, and quality documents for future review.	Quality Department	Continuous	Quality Documentation File

Monitoring Plan

To maintain the effectiveness of the proposed improvements, the following monitoring activities are recommended:

- Inspection checklists should be reviewed during every shift to verify compliance with the inspection procedure.
- Supervisor audits should be conducted regularly to evaluate adherence to visual inspection standards and operating procedures.
- Defect tracking sheets should be updated daily to identify recurring defect patterns and monitor defect escape trends.
- Key Performance Indicators (KPIs), including First Pass Yield (FPY), Defect Escape Rate, and Defects Per Million Opportunities (DPMO), should be reviewed monthly to evaluate process performance.
- Corrective actions should be documented, assigned to responsible personnel, and reviewed until closure.
- Training records should be maintained to ensure that inspectors receive periodic competency development and refresher training.
- Inspection documents should be reviewed periodically to ensure that all procedures and visual standards remain current and applicable.

Key Performance Indicators (KPIs)

The following KPIs are recommended for monitoring the effectiveness of the casting inspection process.

KPI	**Purpose**	**Review Frequency**
Defect Escape Rate	Monitor the percentage of irreparable casting defects reaching downstream operations.	Monthly
First Pass Yield (FPY)	Evaluate the percentage of components passing through the process without permanent rejection.	Monthly
Defects Per Million Opportunities (DPMO)	Measure overall inspection process performance using Six Sigma methodology.	Monthly
Number of Defect Escapes	Track recurring inspection failures and identify improvement opportunities.	Weekly

SOP Compliance	Verify adherence to standardized inspection procedures.	Weekly
Training Completion	Ensure all inspectors receive the required training.	Quarterly

Expected Long-Term Benefits

If the proposed control measures are followed after future implementation, the organization may achieve the following long-term benefits:

- Consistent application of standardized inspection procedures.
- Improved detection of irreparable casting defects during the initial inspection stage.
- Reduced recurrence of defect escapes into downstream operations.
- Better utilization of powder coating and machining resources.
- Improved accountability through regular supervisory audits and documented corrective actions.
- Better communication between casting inspection and final inspection departments.
- Continuous monitoring of process performance using measurable quality indicators.
- Increased operator competency through structured training and periodic performance reviews.
- Sustained process improvement through systematic documentation and continuous review of defect trends.

CONCLUSION

This case study applied the DMAIC methodology to evaluate the effectiveness of the casting inspection process for preventing irreparable casting defects from reaching downstream operations.

Production data collected over a 15-day observation period showed that **189 defective components** escaped the casting inspection stage out of **9,000** processed components, resulting in a **2.10%** defect escape rate, **97.90%** First Pass Yield (FPY), and **21,000 DPMO**. These defect escapes led to an estimated **₹3,591** in downstream processing loss and approximately **10.24 hours** of wasted machining capacity during the observation period.

The analysis identified that the primary causes of defect escapes were related to inspection practices rather than manufacturing processes. The major contributing factors included the absence of standardized inspection procedures, unclear acceptance and rejection criteria, insufficient operator training, greater Focus on inspection quantity than inspection quality, limited supervisory monitoring, and inadequate communication between inspection stages.

Based on these findings, practical recommendations were proposed, including standardized visual inspection criteria, inspection checklists, Standard Operating Procedures (SOPs), structured operator training, supervisor audits, KPI monitoring, and systematic defect tracking. As data collection was conducted during a limited-duration internship without process-ownership authority, these recommendations are presented as a ready to implement action plan for the organization rather than as implemented changes.

Overall, this case study demonstrates how the DMAIC methodology can be used to systematically identify process weaknesses and develop practical improvement strategies to strengthen inspection effectiveness, reduce downstream processing waste, and support continuous quality improvement in manufacturing operations.